

$$\begin{aligned}
& c_2 c_\gamma^{-2} (g_1^2 + g_2^2) + \\
& \frac{1}{16 \pi^2} \left(\frac{1}{48} \frac{1}{m_2} \left(3 g_1^4 \left(2 m_2^2 (-2 + c_2 \gamma^2) + C_H^2 S_4 \gamma^2 \right) 6 g_1^2 g_2^2 \left(4 m_2^2 (-1 + c_2 \gamma^2) + C_H^2 S_4 \gamma^2 \right) + \right. \right. \\
& \quad \left. g_2^4 \left(3 C_H^2 S_4 \gamma^2 + 2 m_2^2 (-18 + c_2 \gamma^2 (4 + 9 c_2^2 + 9 S_2^2)) \right) \right) + \\
& \quad \frac{4}{9} C_H^2 g_2^4 L F_{3,0}[m_2] + \frac{2}{3} C_H^2 g_2^4 L F_{4,-1}[m_2] - \frac{32}{45} C_H^2 g_2^4 L F_{5,-2}[m_2] + \\
& \quad \frac{1}{4} \sum S_2 \gamma S_4 \gamma g_1^2 \frac{1}{m_4^2} (g_1^2 + g_2^2) (C_H^2 - m_2^2) L F_{1,0}[m_d^P] - \frac{1}{12} \sum g_1^4 c_2 \gamma^2 L F_{2,0}[m_d^P] + \\
& \quad \frac{3}{4} S_2 \gamma S_4 \gamma \frac{1}{m_2} \overline{y_d^P} y_d^P (g_1^2 + g_2^2) (-C_H^2 + m_2^2) L F_{1,0}[m_d^P] + c_2 \gamma g_1^2 c_2 \gamma^2 \overline{y_d^P} y_d^P L F_{2,0}[m_d^P] + \\
& \quad \frac{1}{4} \sum S_2 \gamma S_4 \gamma g_1^2 \frac{1}{m_4^2} (g_1^2 + g_2^2) (C_H^2 - m_2^2) L F_{1,0}[m_e^P] - \frac{1}{4} \sum g_1^4 c_2 \gamma^2 L F_{2,0}[m_e^P] + \\
& \quad \frac{1}{4} S_2 \gamma S_4 \gamma \frac{1}{m_2} \overline{y_e^P} y_e^P (g_1^2 + g_2^2) (-C_H^2 + m_2^2) L F_{1,0}[m_e^P] + c_2 \gamma g_1^2 c_2 \gamma^2 \overline{y_e^P} y_e^P L F_{2,0}[m_e^P] + \\
& \quad \frac{1}{4} S_2 \gamma S_4 \gamma \frac{1}{m_4^2} (g_1^2 + g_2^2) (-C_H^2 + m_2^2) (\overline{y_e^P} y_e^P + \sum g_1^2) L F_{1,0}[m_l^P] - \\
& \quad \frac{1}{9} c_2 \gamma (4 c_2^2 \overline{y_e^P} y_e^P (g_1^2 - g_2^2) + \sum c_2 \gamma (g_1^4 + g_2^4)) L F_{2,0}[m_l^P] + \\
& \quad \frac{1}{18} C_H^2 g_2^2 (36 c_2 \gamma c_2^2 \overline{y_e^P} y_e^P + \sum g_2^2 (4 - 9 c_2^2)) L F_{3,0}[m_l^P] + \frac{1}{12} C_H^2 g_2^2 \\
& \quad (-24 c_2 \gamma c_2^2 \overline{y_e^P} y_e^P + \sum g_2^2 (-5 + 6 c_2^2)) L F_{4,-1}[m_l^P] + \frac{45}{8} \sum C_H^2 g_2^4 L F_{5,-2}[m_l^P] + \\
& \quad \frac{1}{4} S_2 \gamma S_4 \gamma \frac{1}{m_4^2} (g_1^2 + g_2^2) (C_H^2 - m_2^2) (-3 \overline{y_d^P} y_d^P + 3 \overline{y_u^P} y_u^P + \sum g_1^2) L F_{1,0}[m_q^P] + \\
& \quad \frac{1}{24} c_2 \gamma (12 c_2 \gamma^2 \overline{y_d^P} y_d^P (g_1^2 + 3 g_2^2) + 12 S_2^2 \overline{y_u^P} y_u^P (g_1^2 - 3 g_2^2) - \sum c_2 \gamma (g_1^4 + 9 g_2^4)) \\
& \quad L F_{2,0}[m_q^P] + \frac{1}{6} C_H^2 g_2^2 (36 c_2 \gamma (c_2^2 \overline{y_d^P} y_d^P - S_2^2 \overline{y_u^P} y_u^P) + \sum g_2^2 (4 - 9 c_2^2)) L F_{3,0}[m_q^P] + \\
& \quad \frac{1}{4} C_H^2 g_2^2 (24 C_2 \gamma (-C_2 \gamma^2 \overline{y_d^P} y_d^P + S_2^2 \overline{y_u^P} y_u^P) + \sum g_2^2 (-5 + 6 c_2^2)) L F_{4,-1}[m_q^P] + \\
& \quad \frac{8}{15} \sum C_H^2 g_2^4 L F_{5,-2}[m_q^P] + \frac{1}{2} \sum S_2 \gamma S_4 \gamma g_1^2 \frac{1}{m_4^2} (g_1^2 + g_2^2) (C_H^2 - m_2^2) L F_{1,0}[m_u^P] - \\
& \quad \frac{1}{3} \sum g_1^4 c_2 \gamma^2 L F_{2,0}[m_u^P] + \frac{3}{4} S_2 \gamma S_4 \gamma \frac{1}{m_2} \overline{y_u^P} y_u^P (g_1^2 + g_2^2) (C_H^2 - m_2^2) L F_{1,0}[m_u^P] - \\
& \quad 2 c_2 \gamma g_1^2 S_2^2 \overline{y_u^P} y_u^P L F_{2,0}[m_u^P] + \frac{1}{16} \frac{1}{m_4^2} S_4 \gamma^2 (C_H^2 - m_2^2) L F_{1,0}[m_a] (g_1^2 + g_2^2)^2 - \\
& \quad \frac{3}{16} S_4 \gamma^2 L F_{1,1}[m_a] (g_1^2 + g_2^2)^2 + \frac{3}{16} C_H^2 S_4 \gamma^2 L F_{1,2}[m_a] (g_1^2 + g_2^2)^2 + \\
& \quad \frac{1}{64} (2 g_1^2 g_2^2 (3 + c_4 \gamma (2 + 5 c_4 \gamma) - 4 S_2 \gamma^4) - g_2^4 (9 + c_4 \gamma (-6 + 5 c_4 \gamma) + 4 S_2 \gamma^4) - \\
& \quad g_2^4 (1 + c_4 \gamma (2 + 5 c_4 \gamma) + 4 S_2 \gamma^4)) L F_{2,0}[m_a] + \frac{1}{288} C_H^2 (-9 g_1^4 ((-1 + c_4 \gamma)^2 + 4 S_2 \gamma^4) - \\
& \quad 18 g_1^2 g_2^2 (-3 + c_4 \gamma (2 + c_4 \gamma) + 4 S_2 \gamma^4) - g_2^4 (17 + 9 c_4 \gamma (6 + c_4 \gamma) + 36 S_2 \gamma^4)) L F_{3,0}[m_a] + \\
& \quad \frac{1}{96} C_H^2 (3 g_1^4 ((-1 + c_4 \gamma)^2 + 4 S_2 \gamma^4) + 6 g_1^2 g_2^2 (-3 + c_4 \gamma (2 + c_4 \gamma) + 4 S_2 \gamma^4) + \\
& \quad g_2^4 (-13 + 3 c_4 \gamma (6 + c_4 \gamma) + 12 S_2 \gamma^4)) L F_{4,-1}[m_a] + \\
& \quad \frac{45}{8} C_H^2 g_2^4 L F_{5,-2}[m_a] + \frac{2}{9} C_H^2 g_2^4 L F_{3,0}[\tilde{\mu}] + \frac{1}{3} C_H^2 g_2^4 L F_{4,-1}[\tilde{\mu}] - \frac{45}{48} C_H^2 g_2^4 L F_{5,-2}[\tilde{\mu}] + \\
& \quad \frac{1}{2} m_1 S_4 \gamma \tilde{\mu} g_1^2 \frac{1}{m_4^2} (g_1^2 + g_2^2) (-C_H^2 + m_2^2) (c_\gamma^2 - S_\gamma^2) L F_{1,1,0}[m_1, \tilde{\mu}] + \frac{1}{2} g_1^4 L F_{2,2,-2}[m_1, \tilde{\mu}] + \\
& \quad \frac{1}{2} g_1^4 (m_1^2 + 8 m_1 S_\gamma \tilde{\mu} c_\gamma + 4 S_\gamma^2 \tilde{\mu}^2 c_\gamma^2) L F_{2,2,-1}[m_1, \tilde{\mu}] + 2 g_1^4 m_1^2 S_\gamma^2 \tilde{\mu}^2 c_\gamma^2 L F_{2,2,0}[m_1, \tilde{\mu}] + \\
& \quad \frac{3}{2} m_2 S_4 \gamma \tilde{\mu} g_2^2 \frac{1}{m_4^2} (g_1^2 + g_2^2) (-C_H^2 + m_2^2) (c_\gamma^2 - S_\gamma^2) L F_{1,1,0}[m_2, \tilde{\mu}] - \frac{8}{3} C_H^2 g_2^2 L F_{2,1,0}[m_2, \tilde{\mu}] + \\
& \quad \frac{1}{2} g_2^4 (5 c_\gamma^4 + 2 S_\gamma^2 c_\gamma^2 + 5 S_\gamma^4) L F_{2,2,-2}[m_2, \tilde{\mu}] + \frac{1}{6} g_2^4 (3 m_2^2 (c_\gamma^4 + 10 S_\gamma^2 c_\gamma^2 + S_\gamma^4) - \\
& \quad 2 C_H^2 (9 c_\gamma^4 + 8 S_\gamma^2 + 9 S_\gamma^4 + c_\gamma^2 (8 - 24 S_\gamma^2)) - 72 m_2 S_\gamma \tilde{\mu} c_\gamma + 36 S_\gamma^2 \tilde{\mu}^2 c_\gamma^2) L F_{2,2,-1}[m_2, \tilde{\mu}] - \\
& \quad \frac{2}{3} m_2 S_\gamma c_\gamma g_2^4 (-9 m_2 S_\gamma c_\gamma \tilde{\mu}^2 + C_H^2 (6 m_2 S_\gamma c_\gamma + 8 \tilde{\mu})) L F_{2,2,0}[m_2, \tilde{\mu}] + \\
& \quad \frac{4}{3} C_H^2 g_2^4 L F_{3,1,-1}[m_2, \tilde{\mu}] + \frac{4}{3} C_H^2 g_2^4 (6 c_\gamma^4 + S_\gamma^2 + 6 S_\gamma^4 + c_\gamma^2 (1 - 12 S_\gamma^2)) L F_{3,2,-2}[m_2, \tilde{\mu}] + \\
& \quad \frac{2}{3} m_2 C_H^2 g_2^4 (-9 m_2 (c_\gamma^2 - S_\gamma^2)^2 + 4 S_\gamma \tilde{\mu} c_\gamma^2) L F_{3,2,-1}[m_2, \tilde{\mu}] - \\
& \quad 4 C_H^2 g_2^4 L F_{3,3,-3}[m_2, \tilde{\mu}] (c_\gamma^2 - S_\gamma^2)^2 + 4 C_H^2 g_2^4 m_2^2 L F_{3,3,-2}[m_2, \tilde{\mu}] (c_\gamma^2 - S_\gamma^2)^2 - \\
& \quad 4 C_H^2 g_2^4 L F_{4,2,-3}[m_2, \tilde{\mu}] (c_\gamma^2 - S_\gamma^2)^2 + 4 C_H^2 g_2^4 m_2^2 L F_{4,2,-2}[m_2, \tilde{\mu}] (c_\gamma^2 - S_\gamma^2)^2 - \\
& \quad 3 c_\gamma^4 \overline{y_d^P} \overline{y_d^S} y_d^P y_d^S L F_{1,1,0}[m_d^P, m_d^T] + \frac{3}{4} S_4 \gamma \frac{1}{m_4^2} (g_1^2 + g_2^2) (C_H^2 - m_2^2) \\
& \quad (\tilde{\mu} \overline{y_d^P} (a_d^P (-c_\gamma^2 + S_\gamma^2) + 2 S_\gamma \tilde{\mu} c_\gamma y_d^P) + \overline{a_d^P} (-2 S_\gamma c_\gamma a_d^P + \tilde{\mu} y_d^P (-c_\gamma^2 + S_\gamma^2))) \\
& \quad L F_{1,1,0}[m_d^P, m_q^P] + c_2 \gamma g_1^2 (c_\gamma \overline{a_d^P} - S_\gamma \tilde{\mu} \overline{y_d^P}) (c_\gamma a_d^P - S_\gamma \tilde{\mu} y_d^P) L F_{2,1,0}[m_d^P, m_q^P] + \\
& \quad C_H^2 c_2 \gamma g_1^2 (c_\gamma \overline{a_d^P} - S_\gamma \tilde{\mu} \overline{y_d^P}) (c_\gamma a_d^P - S_\gamma \tilde{\mu} y_d^P) L F_{2,2,0}[m_d^P, m_q^P] - \\
& \quad c_\gamma^4 \overline{y_e^P} \overline{y_e^S} y_e^P y_e^S L F_{1,1,0}[m_e^P, m_e^T] + \frac{1}{4} S_4 \gamma \frac{1}{m_4^2} (g_1^2 + g_2^2) (C_H^2 - m_2^2) \\
& \quad (\tilde{\mu} \overline{y_e^P} (a_e^P (-c_\gamma^2 + S_\gamma^2) + 2 S_\gamma \tilde{\mu} c_\gamma y_e^P) + \overline{a_e^P} (-2 S_\gamma c_\gamma a_e^P + \tilde{\mu} y_e^P (-c_\gamma^2 + S_\gamma^2))) \\
& \quad L F_{1,1,0}[m_e^P, m_l^P] + c_2 \gamma g_1^2 (c_\gamma \overline{a_e^P} - S_\gamma \tilde{\mu} \overline{y_e^P}) (c_\gamma a_e^P - S_\gamma \tilde{\mu} y_e^P) L F_{2,1,0}[m_e^P, m_l^P] + \\
& \quad C_H^2 c_2 \gamma g_1^2 (c_\gamma \overline{a_e^P} - S_\gamma \tilde{\mu} \overline{y_e^P}) (c_\gamma a_e^P - S_\gamma \tilde{\mu} y_e^P) L F_{2,2,0}[m_e^P, m_l^P] + \frac{1}{4} \frac{1}{m_4^2} (\tilde{\mu} \overline{y_e^P} \\
& \quad (a_e^P (-2 S_\gamma c_2 \gamma c_\gamma m_2^2 (g_1^2 (-1 + c_2 \gamma) + g_2^2 (1 + c_2 \gamma)) + S_4 \gamma C_H^2 (g_1^2 + g_2^2) (c_\gamma^2 - S_\gamma^2)) + \\
& \quad 2 S_\gamma \gamma y_e^P (-S_4 \gamma C_H^2 c_\gamma (g_1^2 + g_2^2) + S_\gamma c_2 \gamma m_2^2 (g_1^2 (-1 + c_2 \gamma) + g_2^2 (1 + c_2 \gamma)))) + \\
& \quad \overline{a_e^P} (2 c_\gamma a_e^P (c_2 \gamma c_\gamma m_2^2 (g_1^2 (-1 + c_2 \gamma) + g_2^2 (1 + c_2 \gamma)) + S_4 \gamma S_\gamma C_H^2 (g_1^2 + g_2^2)) + \tilde{\mu} y_e^P \\
& \quad (-2 S_\gamma c_2 \gamma c_\gamma m_2^2 (g_1^2 (-1 + c_2 \gamma) + g_2^2 (1 + c_2 \gamma)) + S_4 \gamma C_H^2 (g_1^2 + g_2^2) (c_\gamma^2 - S_\gamma^2)))) \\
& \quad L F_{2,1,0}[m_l^P, m_e^P] + \frac{1}{4} \frac{1}{m_4^2} (g_1^2 + g_2^2) (-2 c_\gamma \overline{a_e^P} a_e^P (c_\gamma m_2^2 c_2 \gamma^2 + S_4 \gamma S_\gamma C_H^2) - \\
& \quad \tilde{\mu} (-2 S_\gamma c_\gamma m_2^2 c_2 \gamma^2 + S_4 \gamma C_H^2 (c_\gamma^2 - S_\gamma^2)) (\overline{y_e^P} a_e^P + y_e^P \overline{a_e^P}) + \\
& \quad 2 S_\gamma \tilde{\mu}^2 \overline{y_e^P} y_e^P (S_4 \gamma C_H^2 c_\gamma - S_\gamma m_2^2 c_2 \gamma^2)) L F_{3,1,-1}[m_l^P, m_e^P] + C_H^2 c_2 \gamma \\
& \quad (g_1^2 (-1 + c_2 \gamma) + g_2^2 (1 + c_2 \gamma)) (c_\gamma \overline{a_e^P} - S_\gamma \tilde{\mu} \overline{y_e^P}) (c_\gamma a_e^P - S_\gamma \tilde{\mu} y_e^P) L F_{3,1,0}[m_l^P, m_e^P] - \\
& \quad C_H^2 c_2 \gamma g_1^2 (c_\gamma \overline{a_e^P} - S_\gamma \tilde{\mu} \overline{y_e^P}) (c_\gamma a_e^P - S_\gamma \tilde{\mu} y_e^P) L F_{3,2,-1}[m_l^P, m_e^P] - \frac{1}{2} C_H^2 (-1 + 2 c_2 \gamma) \\
& \quad (2 g_2^2 + 3 c_2 \gamma (g_1^2 + g_2^2)) (c_\gamma \overline{a_e^P} - S_\gamma \tilde{\mu} \overline{y_e^P}) (c_\gamma a_e^P - S_\gamma \tilde{\mu} y_e^P) L F_{4,1,-1}[m_l^P, m_e^P] + \\
& \quad \frac{2}{3} C_H^2 (-2 g_2^2 + 3 c_2 \gamma^2 (g_1^2 + g_2^2)) (c_\gamma \overline{a_e^P} - S_\gamma \tilde{\mu} \overline{y_e^P}) (c_\gamma a_e^P - S_\gamma \tilde{\mu} y_e^P) L F_{5,1,-2}[m_l^P, m_e^P] - \\
& \quad c_\gamma^4 \overline{y_e^P} \overline{y_e^S} y_e^P y_e^S L F_{1,1,0}[m_l^P, m_l^S] - 2 C_H^2 c_\gamma^4 \overline{y_e^P} \overline{y_e^S} y_e^P y_e^S L F_{2,1,0}[m_l^P, m_l^S] + \\
& \quad 2 C_H^2 c_\gamma^4 \overline{y_e^P} \overline{y_e^S} y_e^P y_e^S L F_{3,1,-1}[m_l^P, m_l^S] + \\
& \quad \frac{1}{4} \frac{1}{m_4^2} (\tilde{\mu} \overline{y_d^P} (a_d^P (-2 S_\gamma c_2 \gamma c_\gamma m_2^2 (g_1^2 (1 + 3 c_2 \gamma) + 3 g_2^2 (1 + c_2 \gamma)) + \\
& \quad 3 S_4 \gamma C_H^2 (g_1^2 + g_2^2) (c_\gamma^2 - S_\gamma^2)) + \\
& \quad 2 S_\gamma \tilde{\mu} y_d^P (-3 S_4 \gamma C_H^2 c_\gamma (g_1^2 + g_2^2) + S_\gamma c_2 \gamma m_2^2 (g_1^2 (1 + 3 c_2 \gamma) + 3 g_2^2 (1 + c_2 \gamma)))) + \\
& \quad \overline{a_d^P} (2 c_\gamma a_d^P (c_2 \gamma c_\gamma m_2^2 (g_1^2 (1 + 3 c_2 \gamma) + 3 g_2^2 (1 + c_2 \gamma)) + 3 S_4 \gamma S_\gamma C_H^2 (g_1^2 + g_2^2)) + \\
& \quad \tilde{\mu} y_d^P (-2 S_\gamma c_2 \gamma c_\gamma m_2^2 (g_1^2 (1 + 3 c_2 \gamma) + 3 g_2^2 (1 + c_2 \gamma)) + \\
& \quad 3 S_4 \gamma C_H^2 (g_1^2 + g_2^2) (c_\gamma^2 - S_\gamma^2)))) L F_{2,1,0}[m_q^P, m_d^P] + \frac{3}{4} \frac{1}{m_4^2} (g_1^2 + g_2^2) \\
& \quad (-2 c_\gamma \overline{a_d^P} a_d^P (c_\gamma m_2^2 c_2 \gamma^2 + S_4 \gamma S_\gamma C_H^2) - \tilde{\mu} (-2 S_\gamma c_\gamma m_2^2 c_2 \gamma^2 + S_4 \gamma C_H^2 (c_\gamma^2 - S_\gamma^2)) \\
& \quad (\overline{y_d^P} a_d^P + y_d^P \overline{a_d^P}) - 2 S_\gamma \tilde{\mu}^2 \overline{y_d^P} y_d^P (S_4 \gamma C_H^2 c_\gamma - S_\gamma m_2^2 c_2 \gamma^2)) L F_{3,1,-1}[m_q^P, m_d^P] + \\
& \quad C_H^2 c_2 \gamma (g_1^2 (1 + 3 c_2 \gamma) + 3 g_2^2 (1 + c_2 \gamma)) (c_\gamma \overline{a_d^P} - S_\gamma \tilde{\mu} \overline{y_d^P}) (c_\gamma a_d^P - S_\gamma \tilde{\mu} y_d^P) \\
& \quad L F_{3,1,0}[m_q^P, m_d^P] - C_H^2 c_2 \gamma g_1^2 (c_\gamma \overline{a_d^P} - S_\gamma \tilde{\mu} \overline{y_d^P}) (c_\gamma a_d^P - S_\gamma \tilde{\mu} y_d^P) L F_{3,2,-1}[m_q^P, m_d^P] - \\
& \quad \frac{3}{2} C_H^2 (-2 g_2^2 + c_2 \gamma (1 + 6 c_2 \gamma) (g_1^2 + g_2^2)) (c_\gamma \overline{a_d^P} - S_\gamma \tilde{\mu} \overline{y_d^P}) \\
& \quad (c_\gamma a_d^P - S_\gamma \tilde{\mu} y_d^P) L F_{4,1,-1}[m_q^P, m_d^P] + \\
& \quad 2 C_H^2 (-2 g_2^2 + 3 c_2 \gamma^2 (g_1^2 + g_2^2)) (c_\gamma \overline{a_d^P} - S_\gamma \tilde{\mu} \overline{y_d^P}) (c_\gamma a_d^P - S_\gamma \tilde{\mu} y_d^P) L F_{5,1,-2}[m_q^P, m_d^P] - \\
& \quad 3 (c_\gamma^4 \overline{y_d^P} \overline{y_d^S} y_d^P y_d^S + S_4^2 \overline{y_u^P} \overline{y_u^S} y_u^P y_u^S) L F_{1,1,0}[m_q^P, m_q^S] - \\
& \quad 6 C_H^2 (c_\gamma^2 \overline{y_d^P} y_d^S (c_\gamma^2 \overline{y_d^S} y_d^P - S_2^2 \overline{y_u^S} y_u^P) + S_4^2 \overline{y_u^P} \overline{y_u^S} y_u^P y_u^S) L F_{2,1,0}[m_q^P, m_q^S] + \\
& \quad 6 C_H^2 (c_\gamma^2 \overline{y_d^P} y_d^S (c_\gamma^2 \overline{y_d^S} y_d^P - S_2^2 \overline{y_u^S} y_u^P) + S_4^2 \overline{y_u^P} \overline{y_u^S} y_u^P y_u^S) L F_{3,1,-1}[m_q^P, m_q^S] + \\
& \quad \frac{3}{4} S_4 \gamma \frac{1}{m_4^2} (g_1^2 + g_2^2) (C_H^2 - m_2^2) (\tilde{\mu} \overline{y_u^P} (a_u^P (-c_\gamma^2 + S_\gamma^2) - 2 S_\gamma \tilde{\mu} c_\gamma y_u^P) + \\
& \quad \overline{a_u^P} (2 S_\gamma c_\gamma a_u^P + \tilde{\mu} y_u^P (-c_\gamma^2 + S_\gamma^2))) L F_{1,1,0}[m_q^P, m_u^P] + \\
& \quad \frac{1}{2} c_2 \gamma (g_1^2 - 3 g_2^2) (S_\gamma \overline{y_u^P} - \tilde{\mu} c_\gamma \overline{y_u^P}) (S_\gamma a_u^P - \tilde{\mu} c_\gamma y_u^P) L F_{2,1,0}[m_q^P, m_u^P] + \\
& \quad \frac{1}{2} C_H^2 (c_2 \gamma + g_2^2 (4 - 3 c_2 \gamma)) (S_\gamma \overline{a_u^P} - \tilde{\mu} c_\gamma \overline{y_u^P}) (S_\gamma a_u^P - \tilde{\mu} c_\gamma y_u^P) L F_{2,2,0}[m_q^P, m_u^P] - \\
& \quad 6 C_H^2 c_2 \gamma g_2^2 (S_\gamma \overline{a_u^P} - \tilde{\mu} c_\gamma \overline{y_u^P}) (S_\gamma a_u^P - \tilde{\mu} c_\gamma y_u^P) L F_{3,1,0}[m_q^P, m_u^P] + \\
& \quad C_H^2 g_2^2 (-1 + 3 c_2 \gamma) (S_\gamma \overline{a_u^P} - \tilde{\mu} c_\gamma \overline{y_u^P}) (S_\gamma a_u^P - \tilde{\mu} c_\gamma y_u^P) L F_{3,2,-1}[m_q^P, m_u^P] + \\
& \quad 6 C_H^2 c_2 \gamma g_2^2 (S_\gamma \overline{a_u^P} - \tilde{\mu} c_\gamma \overline{y_u^P}) (S_\gamma a_u^P - \tilde{\mu} c_\gamma y_u^P) L F_{4,1,-1}[m_q^P, m_u^P] + \\
& \quad 6 C_H^2 S_\gamma^2 c_\gamma^2 \overline{y_d^P} y_d^S \overline{y_u^S} y_u^P L F_{2,1,0}[m_q^S, m_q^P] - \\
& \quad 6 C_H^2 S_\gamma^2 c_\gamma^2 \overline{y_d^P} y_d^S \overline{y_u^S} y_u^P L F_{3,1,-1}[m_q^S, m_q^P] + \frac{1}{4} \frac{1}{m_4^2} (\tilde{\mu} \overline{y_u^P} \\
& \quad (a_u^P (-2 S_\gamma c_2 \gamma c_\gamma m_2^2 (g_1^2 (-4 + 3 c_2 \gamma) + 3 c_2 \gamma g_2^2) + 3 S_4 \gamma C_H^2 (g_1^2 + g_2^2) (c_\gamma^2 - S_\gamma^2)) + \\
& \quad 2 \tilde{\mu} c_\gamma y_u^P (c_2 \gamma c_\gamma m_2^2 (g_1^2 (-4 + 3 c_2 \gamma) + 3 c_2 \gamma g_2^2) + 3 S_4 \gamma S_\gamma C_H^2 (g_1^2 + g_2^2))) + \\
& \quad \overline{a_u^P} (2 S_\gamma a_u^P (-3 S_4 \gamma C_H^2 c_\gamma (g_1^2 + g_2^2) + S_\gamma c_2 \gamma m_2^2 (g_1^2 (-4 + 3 c_2 \gamma) + 3 c_2 \gamma g_2^2)) + \tilde{\mu} y_u^P \\
& \quad (-2 S_\gamma c_2 \gamma c_\gamma m_2^2 (g_1^2 (-4 + 3 c_2 \gamma) + 3 c_2 \gamma g_2^2) + 3 S_4 \gamma C_H^2 (g_1^2 + g_2^2) (c_\gamma^2 - S_\gamma^2)))) \\
& \quad L F_{2,1,0}[m_u^P, m_q^P] + \frac{3}{4} \frac{1}{m_4^2} (g_1^2 + g_2^2) (2 S_\gamma \overline{a_u^P} a_u^P (S_4 \gamma C_H^2 c_\gamma - S_\gamma m_2^2 c_2 \gamma^2) + \\
& \quad \tilde{\mu} (2 S_\gamma c_\gamma m_2^2 c_2 \gamma^2 + S_4 \gamma C_H^2 (-c_\gamma^2 + S_\gamma^2)) (\overline{y_u^P} a_u^P + y_u^P \overline{a_u^P}) - \\
& \quad 2 c_\gamma \tilde{\mu}^2 \overline{y_u^P} y_u^P (c_\gamma m_2^2 c_2 \gamma^2 + S_4 \gamma S_\gamma C_H^2)) L F_{3,1,-1}[m_u^P, m_q^P] + \\
& \quad C_H^2 (c_2 \gamma g_1^2 (-4 + 3 c_2 \gamma) + g_2^2 (-2 + 3 c_2 \gamma^2)) (S_\gamma \overline{a_u^P} - \tilde{\mu} c_\gamma \overline{y_u^P}) \\
& \quad (S_\gamma a_u^P - \tilde{\mu} c_\gamma y_u^P) L F_{3,1,0}[m_u^P, m_q^P] - \\
& \quad \frac{1}{2} C_H^2 (c_2 \gamma g_1^2 + g_2^2 (4 - 3 c_2 \gamma)) (S_\gamma \overline{a_u^P} - \tilde{\mu} c_\gamma \overline{y_u^P}) (S_\gamma a_u^P - \tilde{\mu} c_\gamma y_u^P) L F_{3,2,-1}[m_u^P, m_q^P] - \\
& \quad 3 C_H^2 (c_2 \gamma g_1^2 (-2 + 3 c_2 \gamma) + g_2^2 (-2 + 3 c_2 \gamma^2)) \\
& \quad (S_\gamma \overline{a_u^P} - \tilde{\mu} c_\gamma \overline{y_u^P}) (S_\gamma a_u^P - \tilde{\mu} c_\gamma y_u^P) L F_{4,1,-1}[m_u^P, m_q^P] + \\
& \quad 2 C_H^2 (-2 g_2^2 + 3 c_2 \gamma^2 (g_1^2 + g_2^2)) (S_\gamma \overline{a_u^P} - \tilde{\mu} c_\gamma \overline{y_u^P}) (S_\gamma a_u^P - \tilde{\mu} c_\gamma y_u^P) L F_{5,1,-2}[m_u^P, m_q^P] - \\
& \quad 3 S_\gamma^2 \overline{y_u^P} \overline{y_u^S} y_u^P y_u^S L F_{1,1,0}[m_u^P, m_u^S] - \frac{1}{4} C_H^2 c_2 \gamma^2 (g_1^2 + g_2^2) L F_{2,1,-1}[\tilde{\mu}, m_1] + \\
& \quad \frac{1}{2} m_1 \tilde{\mu} g_1^2 \frac{1}{m_2} (g_1^2 + g_2^2) (-2 S_\gamma S_\gamma m_2^2 c_2 \gamma^2 + S_4 \gamma C_H^2 (c_\gamma^2 - S_\gamma^2)) L F_{2,1,0}[\tilde{\mu}, m_1] + \\
& \quad \frac{1}{2} g_1^2 c_2 \gamma^2 (g_1^2 + g_2^2) L F_{3,1,-2}[\tilde{\mu}, m_1] + \frac{1}{2} g_1^2 \frac{1}{m_2} (g_1^2 + g_2^2) \\
& \quad (m_1 S_4 \gamma C_H^2 \tilde{\mu} (-c_\gamma^2 + S_\gamma^2) + 2 m_2^2 c_2 \gamma^2 (-2 C_H^2 + m_1 S_\gamma \tilde{\mu} c_\gamma)) L F_{3,1,-1}[\tilde{\mu}, m_1] - \\
& \quad 2 m_1 S_\gamma C_H^2 \tilde{\mu} c_\gamma g_1^2 c_2 \gamma^2 (g_1^2 + g_2^2) L F_{3,1,0}[\tilde{\mu}, m_1] + \frac{3}{2} C_H^2 g_1^4 L F_{3,2,-2}[\tilde{\mu}, m_1] + \\
& \quad \frac{1}{2} C_H^2 g_1^4 (3 m_1^2 + 20 m_1 S_\gamma \tilde{\mu} c_\gamma + 8 S_\gamma^2 \tilde{\mu}^2 c_\gamma^2) L F_{3,2,-1}[\tilde{\mu}, m_1] + \\
& \quad 4 C_H^2 g_1^4 m_1^2 S_\gamma^2 \tilde{\mu}^2 c_\gamma^2 L F_{3,2,0}[\tilde{\mu}, m_1] + \\
& \quad \frac{1}{3} C_H^2 g_1^2 (-5 g_2^2 + 12 c_2 \gamma^2 (g_1^2 + g_2^2)) L F_{4,1,-2}[\tilde{\mu}, m_1] + \\
& \quad 2 m_1 S_\gamma C_H^2 \tilde{\mu} c_\gamma g_1^2 (-g_2^2 + 3 c_2 \gamma^2 (g_1^2 + g_2^2)) L F_{4,1,-1}[\tilde{\mu}, m_1] - \\
& \quad 4 C_H^2 g_1^4 L F_{4,2,-3}[\tilde{\mu}, m_1] + C_H^2 g_1^4 (-m_1^2 - 8 m_1 S_\gamma \tilde{\mu} c_\gamma - 4 S_\gamma^2 \tilde{\mu}^2 c_\gamma^2) L F_{4,2,-2}[\tilde{\mu}, m_1] - \\
& \quad C_H^2 g_1^2 m_1^2 S_\gamma^2 \tilde{\mu}^2 c_\gamma^2 L F_{4,2,-1}[\tilde{\mu}, m_1] - \\
& \quad \frac{2}{3} C_H^2 g_1^2 (-2 g_2^2 + 3 c_2 \gamma^2 (g_1^2 + g_2^2)) L F_{5,1,-3}[\tilde{\mu}, m_1] - \\
& \quad \frac{4}{3} m_1 S_\gamma C_H^2 \tilde{\mu} c_\gamma g_1^2 (-2 g_2^2 + 3 c_2 \gamma^2 (g_1^2 + g_2^2)) L F_{5,1,-2}[\tilde{\mu}, m_1] - \\
& \quad \frac{9}{4} g_2^2 c_2 \gamma^2 (g_1^2 + g_2^2) L F_{2,1,-1}[\tilde{\mu}, m_2] + \\
& \quad \frac{3}{2} m_2 \tilde{\mu} g_2^2 \frac{1}{m_2} (g_1^2 + g_2^2) (-2 S_\gamma c_\gamma m_2^2 c_2 \gamma^2 + S_4 \gamma C_H^2 (c_\gamma^2 - S_\gamma^2)) L F_{2,1,0}[\tilde{\mu}, m_2] + \\
& \quad \frac{3}{2} g_2^2 c_2 \gamma^2 (g_1^2 + g_2^2) L F_{3,1,-2}[\tilde{\mu}, m_2] + \frac{1}{6} g_2^2 \frac{1}{m_2} (-4 C_H^2 m_2^2 (-8 g_2^2 + 9 c_2 \gamma^2 (g_1^2 + g_2^2)) + \\
& \quad 9 m_2 \tilde{\mu} (g_1^2 + g_2^2) (2 S_\gamma c_\gamma m_2^2 c_2 \gamma^2 + S_4 \gamma C_H^2 (-c_\gamma^2 + S_\gamma^2))) L F_{3,1,-1}[\tilde{\mu}, m_2] - \\
& \quad \frac{2}{3} m_2 S_\gamma C_H^2 \tilde{\mu} c_\gamma g_2^2 (-8 g_2^2 + 9 c_2 \gamma^2 (g_1^2 + g_2^2)) L F_{3,1,0}[\tilde{\mu}, m_2] + \\
& \quad \frac{1}{6} C_H^2 g_2^4 (69 c_\gamma^4 + 16 S_\gamma^2 + 69 S_\gamma^4 + c_\gamma^2 (16 - 6 S_\gamma^2)) L F_{3,2,-2}[\tilde{\mu}, m_2] + \\
& \quad \frac{1}{6} C_H^2 g_2^4 (3 m_2^2 (3 c_\gamma^4 + 38 S_\gamma^2 c_\gamma^2 + 3 S_\gamma^4) + 4 m_2 S_\gamma \tilde{\mu} c_\gamma (8 + 51 c_\gamma^2 + 51 S_\gamma^2) + 72 S_\gamma^2 \tilde{\mu}^2 c_\gamma^2) \\
& \quad L F_{3,2,-1}[\tilde{\mu}, m_2] + 12 C_H^2 g_2^4 m_2^2 S_\gamma^2 \tilde{\mu}^2 c_\gamma^2 L F_{3,2,0}[\tilde{\mu}, m_2] + \\
& \quad 3 C_H^2 g_2^2 (-3 g_2^2 + 4 c_2 \gamma^2 (g_1^2 + g_2^2)) L F_{4,1,-2}[\tilde{\mu}, m_2] + \\
& \quad 2 m_2 S_\gamma C_H^2 \tilde{\mu} c_\gamma g_2^2 (-7 g_2^2 + 9 c_2 \gamma^2 (g_1^2 + g_2^2)) L F_{4,1,-1}[\tilde{\mu}, m_2] - \\
& \quad C_H^2 g_2^4 (7 c_\gamma^4 + 2 S_\gamma^2 c_\gamma^2 + 7 S_\gamma^4) L F_{4,2,-3}[\tilde{\mu}, m_2] - \\
& \quad C_H^2 g_2^4 (m_2^2 (c_\gamma^4 + 14 S_\gamma^2 c_\gamma^2 + S_\gamma^4) + 32 m_2 S_\gamma \tilde{\mu} c_\gamma + 16 S_\gamma^2 \tilde{\mu}^2 c_\gamma^2) L F_{4,2,-2}[\tilde{\mu}, m_2] - \\
& \quad 16 C_H^2 g_2^4 m_2^2 S_\gamma^2 \tilde{\mu}^2 c_\gamma^2 L F_{4,2,-1}[\tilde{\mu}, m_2] - 2 C_H^2 g_2^2 (-2 g_2^2 + 3 c_2 \gamma^2 (g_1^2 + g_2^2)) L F_{5,1,-3}[\tilde{\mu}, m_2] - \\
& \quad 4 m_2 S_\gamma C_H^2 \tilde{\mu} c_\gamma g_2^2 (-2 g_2^2 + 3 c_2 \gamma^2 (g_1^2 + g_2^2)) L F_{5,1,-2}[\tilde{\mu}, m_2] + \\
& \quad 8 C_H^2 g_1^2 g_2^2 L F_{2,2,-1,-2}[m_2, \tilde{\mu}] + \\
& \quad 2 C_H^2 g_1^2 g_2^2 (3 m_1 m_2 + 2 S_\gamma \tilde{\mu} c_\gamma (7 m_1 + 5 m_2) + 12 S_\gamma^2 \tilde{\mu}^2 c_\gamma^2) L F_{2,2,1,-1}[m_2, \tilde{\mu}, m_1] + \\
& \quad 16 m_1 m_2 C_H^2 g_1^2 g_2^2 S_\gamma^2 \tilde{\mu}^2 c_\gamma^2 L F_{2,2,1,0}[m_2, \tilde{\mu}, m_1] - 4 C_H^2 g_1^2 g_2^2 L F_{3,2,1,-3}[m_2, \tilde{\mu}, m_1] - \\
& \quad 4 C_H^2 g_1^2 g_2^2 (m_1 m_2 + 4 S_\gamma \tilde{\mu} c_\gamma (m_1 + m_2) + 4 S_\gamma^2 \tilde{\mu}^2 c_\gamma^2) L F_{3,2,1,-2}[m_2, \tilde{\mu}, m_1] - \\
& \quad 16 m_1 m_2 C_H^2 g_1^2 g_2^2 S_\gamma^2 \tilde{\mu}^2 c_\gamma^2 L F_{3,2,1,-1}[m_2, \tilde{\mu}, m_1] + \\
& \quad 6 c_\$$